Jessie Levillain

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My Webpage

Academic career

From Dec. 2024 **Post-doctoral fellow**, *CNES, Campus de la donnée*, Toulouse, France.
Supervision: Nicolas Gasnier. Project title: "Synergistic use of satellite data for monitoring surface water dynamics".

Education

Oct. 2021–Sept. 2024 **PhD in applied mathematics**, *CMAP, École polytechnique*, Palaiseau, France.
Filament models for swimming at the microscopic scale.
Supervision: François Alouges and Aline Lefebvre-Lepot.
Defense: September 27th, 2024.
Reviewers:
- Astrid Decoene, Université de Bordeaux
- Céline Grandmont, INRIA Paris

Examiners:
- Josselin Garnier, École polytechnique
- Laëtitia Giraldi, INRIA Sophia Antipolis
- Sébastien Martin, Université Paris-Cité

2017–2021 **École Normale Supérieure Paris-Saclay diploma**, *ENS Paris-Saclay*, Gif-sur-Yvette, France (*Formerly* ENS Cachan), Civil servant and student.
- M2 Analyse Modélisation Simulation at Université Paris-Saclay, *magna cum laude* (numerical methods, mathematical modeling and parallel computing ; optimization, control and optimal transport).
- Pre-doctoral research year abroad.
- M1 Hadamard at Ecole Normale Supérieure Paris-Saclay, Université Paris-Saclay and École polytechnique, *magna cum laude* (dynamical systems, analysis, geometry, optimization and applications to physics).
- Bachelor's Degree in Mathematics at ENS Paris-Saclay in partnership with Université Paris Diderot, *cum laude* (analysis, algebra, probability theory, quantum mechanics).

2015–2017 **Lycée Marcelin Berthelot**, *CPGE MPSI-MP**, Saint-Maur-des-Fossés, France, *summa cum laude*.
Two intensive years of training in mathematics, physics, and computer science, in order to take competitive examinations and enter the most prestigious engineering or research schools.

Short-term research visits and internships

Oct. 2025 **LS2N, Centrale Nantes**, *Nantes, France*, 1 week.
Invited by C. Moreau.

Oct. 2025 **LMBA, Université de Bretagne Occidentale**, *Brest, France*, 3 days.
Invited by V. Calvez and V. Milisic.

Aug. 2023 **RIMS, Kyoto University**, *Kyoto, Japan*, 1 week.
Invited by C. Moreau and K. Ishimoto.

May – June 2023 **SISSA**, *Trieste, Italy*, 2 months.
Collaboration with I. Anello and A. DeSimone. Funded by 'Programme Visibilité Scientifique Junior', Fondation Mathématiques Hadamard.

Oct. 2022 **RIMS, Kyoto University**, *Kyoto, Japan*, 3 weeks.
Invited by C. Moreau and K. Ishimoto.

April – August 2021 **CMAP, École polytechnique**, *Internship*, Palaiseau, France.
Simple micro-swimmer models. Internship supervised by F. Alouges and A. Lefebvre-Lepot.

2019 – 2020 **Politecnico di Torino**, *Pre-doctoral year of research abroad*, Turin, Italy.
Internship in the Energy department of Politecnico di Torino, supervised by F. Freschi, on "H-matrices and ACA method" as a part of the MICEV project.

April – August 2019 **UCD School of mathematics and statistics**, *Internship*, Dublin, Ireland.
Internship supervised by F. Dias, in the School of Mathematics and Statistics on "Extended Green's Law : Calculating Beta using machine learning".

July 2018 **DTU Energi**, *Internship*, Copenhagen, Denmark.
Internship supervised by K. Kirstein Nielsen, C. Bahl and R. Bjørk, entitled "Optimizing magnet structures with neural networks".

January – June 2018 **CMLA, ENS Paris-Saclay**, *Internship*, Cachan , France.
"Deep burst denoising": Image denoising thanks to neural networks (machine learning). Supervised by J.-M. Morel, P. Arias and T. Ehret, in tandem with R. Zhang.

Teaching

2021 – 2024 **École polytechnique**, *Teaching fellow*, Palaiseau, France.
Course list:
- MAP350: Preparatory mathematics course in French, 1st year of Ingénieur Polytechnicien program
- Computational mathematics (MAA106, Bachelor of Science, 1st year)
- Numerical linear algebra and optimization (MAA251, previously MAA208, Bachelor of Science, 2nd year)

Jan. – June 2021 **Institut Villebon Georges Charpak**, *Orsay, France*.
Tutoring in mathematics for bachelor students.

2017 – 2018 **INSEP**, *Mathematics guest teacher*, Paris, France, Classes for high-schoolers.

2012 – 2015 **Lycée Notre Dame de la Providence**, France.
Tutoring in Mathematics, English and all subjects for middle- and high-schoolers.

Supervision

2026 **Research internship (M2)**, *IECL, Université de Lorraine*, Metz, France.
Co-supervision of A. Schlater (Jena University) with M. D'Achille and P. Moyal on "Random assignment problems for satellite image registration", from April 2026. Duration: 6 months.

2025 **Research internship (M2)**, *LMO, Université Paris-Saclay*, Orsay, France.
Co-supervision of T. Quarck with M. D'Achille on "Random assignment problems for satellite data acquisition". Duration: 6 months.

2025 **Research internship (M2)**, *CNES*, Toulouse, France.
Co-supervision of L. Marsal with N. Gasnier on "Uncertainty quantification and data hybridation for water storage evaluation". Duration: 6 months.

2023 **Bachelor thesis (L3)**, *CMAP, École polytechnique*, Palaiseau, France.
Supervision of A. Failler. "On the Perron-Frobenius theorem: discovering Google's PageRank algorithm". Duration: 2 months.

2022 **Bachelor thesis (L3)**, *CMAP, École polytechnique*, Palaiseau, France.
Supervision of D. Dac on "Introduction to image denoising: on BM3D". Duration: 2 months.

Publications

**Contribution-based author order.*

Journal articles (3)

2025 François Alouges, Aline Lefebvre-Lepot, Jessie Levillain, and Clément Moreau. The N-link model for slender rods in a viscous fluid: well-posedness and convergence to classical elastohydrodynamics equations. *Nonlinearity*, volume 38, page 125017. IOP Publishing, dec 2025.

2025 François Alouges, Irene Anello, Antonio DeSimone, Aline Lefebvre-Lepot, and Jessie Levillain. Some mathematical models for flagellar activation mechanisms. *Mathematical Models and Methods in Applied Sciences*, volume 35, pages 2395–2424, 2025.

2023 François Alouges, Aline Lefebvre-Lepot, and Jessie Levillain. A limiting model for a low reynolds number swimmer with N passive elastic arms. *AIMS Mathematics in Engineering*, volume 5, pages 1–20, 2023.

Conference proceedings (2)

2025 Romain Thoreau, Jessie Levillain, and Dawa Derksen*. Can multimodal representation learning by alignment preserve modality-specific information? *accepted as Workshop paper at MACLEAN - ECML/PKDD, to appear in Springer Lecture Notes in Computer Science*, 2025.

2025 Jessie Levillain, François Alouges, Akash Choudhary, Antonio DeSimone, Sankalp Nambiar, and Ida Bochert*. Bi-directional low reynolds number swimmers with passive elastic arms. *ESAIM:ProcS, CJC-MA 2023*, 2025.

Preprints (1)

2025 Jessie Levillain and Jérôme Monnier*. Bathymetry estimations in river networks from altimetry measurements. *submitted*, 2025.

Communications

National and international conferences

March 2026 **CJC-MA 2026**, *Champs sur Marne, France*, Micro-filament modeling and convergence of an N-link model.

Nov. 2025 **SIAM PD25**, *Pittsburgh, PA, USA*, Two talks.

June 2025 **SMAI 2025**, *Carcans, France*, Talk on bathymetry inversions in river networks.

May 2024 **CANUM 2024**, *Le bois-plage-en-Ré, Ile de Ré, France*, Mini-symposium organizer and talk on low-Reynolds-number swimming.

Sept. 2023 **CJC-MA 2023**, *Gif-sur-Yvette, France*, Bi-directional low-Reynolds-number swimmer models.

Aug. 2023 **ICIAM 2023**, *Tokyo, Japan*, Co-organization and chairing of a mini-symposium on low-Reynolds-number swimming with C. Moreau and talk.

Sept. 2022 **CJC-MA 2022**, *Calais, France*, A mathematical model for flagellar activation mechanisms.

June 2022 **CANUM 2020+2**, *Évian-les-bains, France*, Flagellar Locomotion of Micro-organisms.
invited by L. Giraldi and P. Lissy (Mini-symposium)

Seminars and workshops

Mar. 2026 **Numerical analysis and modeling seminar**, *IRMAR, Université de Rennes, France*.

Feb. 2026 **LMAH Seminar**, *LMAH, Université Le Havre Normandie, France*.

Feb. 2026 **PDE Seminar**, *IECL, Université de Lorraine, Metz, France*.

Jan. 2026 **Young Researcher days**, *X recherche, Maison des X, Paris, France*.

Jan. 2026 **EDPAN Seminar**, *LMBP, Université Clermont Auvergne, Clermont-Ferrand, France*.

Jan. 2026 **Applied analysis team Seminar**, *I2M, Université Aix-Marseille, Marseille, France*.

Nov. 2025 **PDE Seminar**, *Department of Mathematics, Purdue University, Lafayette IN, USA*.

Nov. 2025 **Applied and Computational Math seminar**, *Department of Mathematics, University of Wisconsin-Madison, Madison, WI, USA*.

Oct. 2025 **ReV team seminar**, *LS2N, Centrale Nantes, Nantes, France*.

Oct. 2025 **MACS team seminar**, *LMJL, Université de Nantes, Nantes, France*.

June 2025 **SuPerGRandMa seminar**, *IMO, Université Paris-Saclay, Orsay, France*.

March 2025 **Séminaire "Analyse, Phénomènes Stochastiques et Applications"**, *LMBA, Université de Bretagne Occidentale, Brest, France.*

Oct. 2024 **Workshop "Bio- and bio-inspired locomotion of elongated bodies across scales and fields"**, *LS2N, Ecole Centrale Nantes, Nantes, France.*

April 2024 **Séminaire de l'équipe CALISTO**, *INRIA Université Côte D'Azur, Sophia Antipolis, France.*

March 2024 **Séminaire des doctorants du LAMFA**, *LAMFA, Université d'Amiens, France.*

Aug. 2023 **RIMS Fluid Dynamics Workshop**, *Kyoto, Japan*, Flagellar activation mechanisms: inside an N-axoneme .

June 2023 **AJS Seminars**, *SISSA, Trieste, Italy.*

Jan. 2023 **Séminaire des doctorants du CMAP et du CMLS**, *École polytechnique, Palaiseau, France.*

Oct. 2022 **RIMS Fluid Dynamics Group seminar**, *Kyoto University, Japan.*

Oct. 2022 **Microscale Ocean Biophysics online seminar**.

Poster presentations

Nov. 2025 **JMBS**, *Université Montpellier 2*, Montpellier, France.

Oct. 2025 **CNES Young Researcher Days**, *CNES*, Toulouse, France.

May 2025 **7e Symposium du GDR MADICS**, *Université Toulouse Capitole*, Toulouse, France.

Sept. 2024 **Conférence des 50 ans du CMAP**, *École polytechnique*, Palaiseau, France.

July 2023 **XX Jacques-Louis Lions Spanish-French School on Numerical Simulations in Physics & Engineering**, *CIMNE*, Barcelona, Spain.

Grants and awards

2025 **CNES project grant**, *CNES*, 15k€.
Creation and successful funding of a project by CNES, in collaboration with N. Gasnier (CNES) and M. D'Achille (IECL). Funds attributed to N. Curien (LMO).

2023 **Vivaldi Program**, *FMJH and LMH*, 1.3k€.
Funded XX Jacques-Louis Lions Spanish-French School on Numerical Simulations in Physics & Engineering.

2023 **Junior scientific visibility**, *FMJH and LMH*, 2.3k€.
Funded research stay at SISSA, Trieste, Italy.

2022 **Energy4Climate**, *Institut Polytechnique de Paris.*
Funded conference CANUM 2020+2.

2021 **EDMH**, *Institut Polytechnique de Paris.*
PhD funding from doctoral school EDMH.

2019 **Civil servant competitive exam**, *ENS Paris-Saclay.*
Laureate of the competitive exam at the end of the first year of Masters to become a civil servant at ENS Paris-Saclay.

2019 **Erasmus+ internship grant**.
Partially funded internship in Dublin.

2018 **Sophie Germain excellence scolarship**, *FMJH*, 10k€ per year, for two years.
Masters scolarship for outstanding students pursuing a degree in any school member of the Paris-Saclay group.

Organization of scientific events

July 2026 **MicroSwim26**, *Centre Henri Lebesgue, Brest*, Co-organization of an international conference on modeling and mathematical analysis of micro-swimmers in biology with Thierry Aubry (IRDL, UBO),Tristan Montier (GGB, UBO) and Vuk Milisic (LMBA, UBO).

March 2026 **CJC-MA 2024**, *Champs sur Marne, France*, Session chair.

May 2024 **CANUM 2024**, *Le bois-plage-en-Ré, Ile de Ré, France*, Mini-symposium organizer on low-Reynolds-number swimming.

Aug. 2023 **ICIAM 2023**, *Tokyo, Japan*, Co-organization and chairing of a mini-symposium on low-Reynolds-number swimming with C. Moreau.

Collective responsibilities and diffusion

– **Reviewer for Kyoto Journal of Mathematics**, One paper.

Apr. 2026 **Scientific Game Jam**, *Cité de l'espace, Toulouse, France*, Created a game on buiolding and controlling a micro-robot for minimally invasive surgery: `https://itch.io/jam/scientific-game-jam-2026/rate/4470675`.

Jan. 2022 – 2024 **PhD student representative at CMAP lab council**, *CMAP, École polytechnique*, Organization of welcome days for PhD students and postdocs at CMAP, and of a mental health day.

Jan. 2024 **Journée Maths en Herbe**, *IHES, Bures-sur-Yvette, France*, Open science day, introductory talk on micro-swimming.

Personal commitments

Since Nov. 2024 **Occit'à bloc**, *Toulouse*.
Bouldering competitor at a national level; involvement in the streering committee.

Since Sept. 2024 **Marie Rose Escalade**, *Fontainebleau, France*, Organizer and one of the co-founders.
Climbing collective for diversity and inclusion.

2022 – 2024 **Women's Bouldering Festival mentor**, *Fontainebleau, France*.

2022 – 2024 **Girls in Bleau coach**, *Fontainebleau, France*.
Coach at local outdoor climbing club for women.

2022 **Online guidebook creation and crag cleanup**, *La Troche, Palaiseau, France*.
P. Lagouanelle , J. Levillain, and A. Grisel-Davy. La Troche guidebook for bleau.info and Cosiroc. `PDF:https://bleau.info/images/topos/la_troche_apr_2022_compressed.pdf`

2017 – 2020 **Involvement in various student associations**, *ENS Paris-Saclay*.
Co-founder and active member of cooking club Bon Viv[ens] (2018-2020), active member of the student union (2017-2018).

2014 – 2018 **United Network EU**, *Committee director*, Rome, Italy and Paris, France.
IMUNA/United Network EU: Planning and chairing debates in English in Rome (IMUN) and Paris (FMUN) for high school students in English in direct partnership with the UN. Speeches and training classes for students on the subjects they will be debating on, and on rules of procedure during debates.

Miscellaneous skills

Programming Languages: Advanced level in MATLAB, Python. Intermediate level in C/C++ and CamL.

Office: Proficient in LaTeX, beamer and office tools.

Languages

English: Cambridge C2 diploma, lived in America.

French: Native language.

Italian: Advanced, B2-C1.

German: Intermediate, Goethe Institute B2 diploma.

Korean: Beginner.